Claire Kim

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PhD Candidate in Computer Science and Applied AI/ML Engineer specializing in LLM systems, inference, evaluation, RAG, and agentic AI. Built and productionized enterprise AI solutions at Amazon, AMD, and Cognizant, spanning intelligent LLM routing, scalable ML pipelines, multi-agent systems, and automated evaluation. Experienced in translating research advances into reliable production AI systems on cloud and GPU infrastructure.

EDUCATION

PhD, Computer Science **2021 – December 2026**
Arizona State University, Tempe, AZ
- Advisors: Dr. Huan Liu and Dr. Mickey Mancenido

M.E., Computer Science & Engineering **2017 – 2019**
B.S., Computer Science & Engineering **2013 – 2017**
Korea University, Seoul, South Korea

WORK EXPERIENCE

Applied AI Engineer — Sep 2026 –
Cognizant **Remote**
- Designed and developed an SLM-based LLM gateway for dynamic request routing across heterogeneous language models based on query characteristics and model capabilities.

Applied Scientist Intern — Sep – Dec 2025
Amazon **Bellevue, WA**
- Improved LSTM-based production sentiment forecasting models supporting community operations at Amazon by developing advanced sequence modeling approaches and improving prediction robustness at scale.
- Designed and productionized an end-to-end ML pipeline spanning data ingestion, model training, experiment tracking, automated deployment, and production monitoring using Python, AWS, MLflow, and CI/CD.

AI/ML Intern — May – Aug 2025
AMD **Austin, TX**
- Architected and deployed Q-RAG, a question-centric RAG framework leveraging synthetic QA generation and LLM-as-a-Judge evaluation to improve enterprise knowledge retrieval.
- Developed automated mechanisms for identifying documentation gaps and hallucination-prone queries, enabling systematic evaluation of engineering support systems.
- Designed scalable AI evaluation infrastructure enabling automated benchmarking of enterprise RAG systems while significantly reducing manual expert review.

Software Development Intern — Aug – Dec 2024
AMD **Austin, TX**
- Designed production AI inference architecture supporting enterprise RAG, multi-agent orchestration, user-feedback integration, and large-scale experimentation.
- Optimized AI inference and data pipelines to improve throughput, reduce evaluation latency, and eliminate performance bottlenecks across scalable production deployments.
- Collaborated with AI engineers and domain experts to rapidly prototype, iterate, and deploy enterprise AI capabilities.

Graduate Research Assistant — May 2022 – Aug 2024
DHS-CAOE (Center for Accelerating Operational Efficiency) **Tempe, AZ**
- Designed NLP solutions for topic modeling and text summarization using BERT and Llama-2.

- Partnered with researchers, domain experts, and software engineers to architect a trustworthy multi-agent RAG system supporting AI-assisted intelligence analysis.
- Delivered an interactive analytics dashboard enabling stakeholders to explore AI-generated insights through web-based visualizations.

Graduate Research Assistant **Jan 2021 – Aug 2022**

ONR (Office of Naval Research Project), Arizona State University **Tempe, AZ**

- Researched the integration and mutual influence of online and offline COVID-19 datasets using topic modeling.
- Analyzed 2M tweets for sentiment and stance detection in pandemic-related discussions.

SELECTED PROJECTS

Adaptive Triggering for Bias Correction in LLM Reasoning — Under Review

- Formulated inference-time bias correction as an online change-point detection problem, using a trigger to decide when a developing LLM reasoning trajectory shows enough evidence of stereotype reliance to warrant correction.
- Designed and evaluated white-box (logit-based) and black-box (LLM-judge) bias-risk signals across seven open-weight models and GPT-4o-mini, running large-scale experiments on ASU's Sol supercomputer.
- Corrected four undocumented schema errors in the BBQ bias benchmark that had left 17–50% of items in affected categories mis-scored, and open-sourced the fix.

TECHNICAL SKILLS

LLM Systems: RAG, Agentic AI, Multi-Agent Systems, LLM Routing, LLM Evaluation, LLM-as-a-Judge, Synthetic Data Generation, Inference-Time Scaling
Training & Alignment: Fine-Tuning, RLHF, RLAIF, LoRA/PEFT, HuggingFace
ML Frameworks: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy
ML Infrastructure: AWS, GCP, Domino, Docker, MLflow, TensorRT, GPU Inference, Model Serving, CI/CD
LLM Frameworks: LangChain, LlamaIndex
Software: Python, SQL, Java, JavaScript, Bash, REST APIs, Git, Linux

SELECTED PUBLICATIONS | Google Scholar

AI Magazine'25 PADTHAI-MM: A Principled Approach for the Design of Trustworthy, Human-Centered AI systems using the MAST Methodology

ASONAM'24 Robust Stance Detection: Understanding Public Perceptions in Social Media

JAIR'23 Evaluating Trustworthiness of AI-Enabled Decision Support Systems: Validation of the Multisource AI Scorecard Table (MAST)

ACADEMIC SERVICE

Reviewer at **AMLC 2025** (Amazon Gen AI Evaluation Workshop)	2025
Invited Reviewer for **NeurIPS 2025** (Reliable ML Workshop)	2025
Program Committee (PC) member of **ASONAM 2024** conference	2024
Invited talk at Machine Learning Day 2024, Arizona State University	2024
Program Committee (PC) member of **ASONAM**, **SBP-BRiMS 2023** conferences	2023
Invited Reviewer for **EMNLP 2023** conference	2023
Reviewer at **ECML-PKDD, ACM MultiMedia, ASONAM,** and **AAAI** conferences	2022